A job ready bootcamp in C++, DSA and IOT

Arrays



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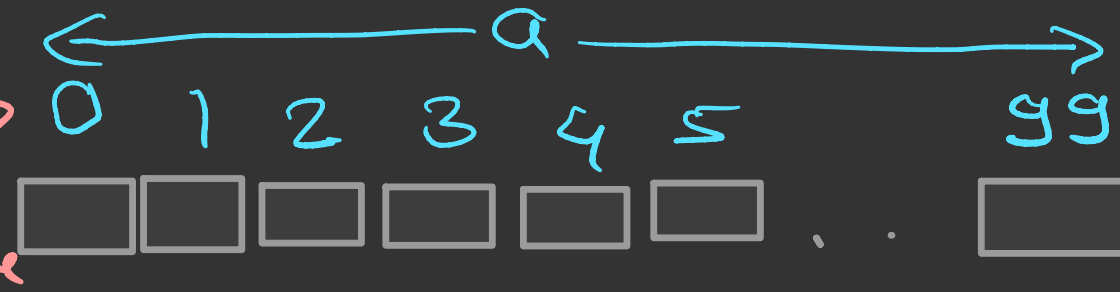
Agenda

1. Introduction to Arrays
2. Array declaration rules
3. Bound Checking
4. Sorting
5. Function call by passing arrays
6. Two dimensional arrays
7. Multi dimensional arrays.

Introduction to Arrays

- Array is a linear collection of similar elements.
- Array is also known as subscript variable

Write a program to calculate average of 100 numbers.




```
int main()
{
    int a[100], i, sum=0;
    float avg;
    printf("Enter 100 nos");
    for(i=0; i<=99; i++)
        scanf("%d",&a[i]);
    for(i=0; i<=99; i++)
        sum = sum + a[i];
    avg = sum/100.0;
    printf("Average is %f", avg);
    return 0;
}
```

a[3] = 30;

[] subscript operator

a[i] x+y 3+4

Array Declaration Rules

① `int a[];` Error

Can't be empty

② `int a[5];`

- Natural number
- Total number of variables in array
- Not an index

③ `int a[5];`



local array when not initialized contains garbage values.

whatever is the size of an array it always consumes memory in a sequential fashion.

④ You can initialize array during declaration

```
int a[5] = {10, 50, 30, 70, 20};
```



⑤ You cannot initialize an array during declaration more than its size

```
int a[5] = {10, 50, 30, 70, 20, 80, 40};
```

Error

(6) You can initialize an array during declaration with lesser values than the size of an array.

```
int a[5] = {10, 50};
```



Remaining variables in array will contain 0. and not garbage value.

⑦ During declaration you can leave [ ] empty only when you initialize array at the same time.

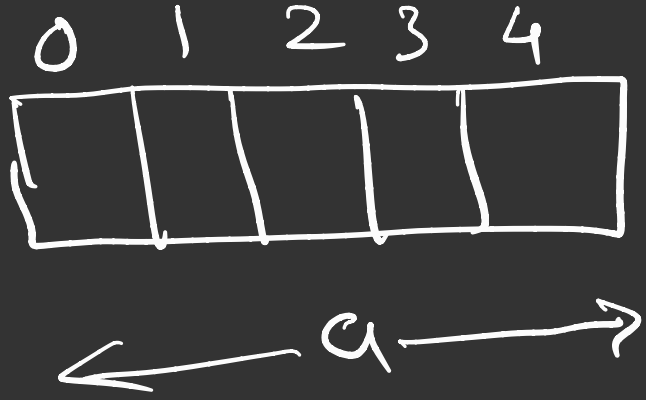
```
int a[] = {10, 50, 30, 80, 20};
```

Bound Checking

```
int a[5] = {10, 20, 50, 90, 30, 60, 70};
```
Error

```
int i, a[5];

for(i=0; i<=9; i++)
    scanf("%d", &a[i]);
```



Sorting

- Arranging elements in some logical order is known as sorting.

- By default, for numbers sorting means arranging elements in ascending order.

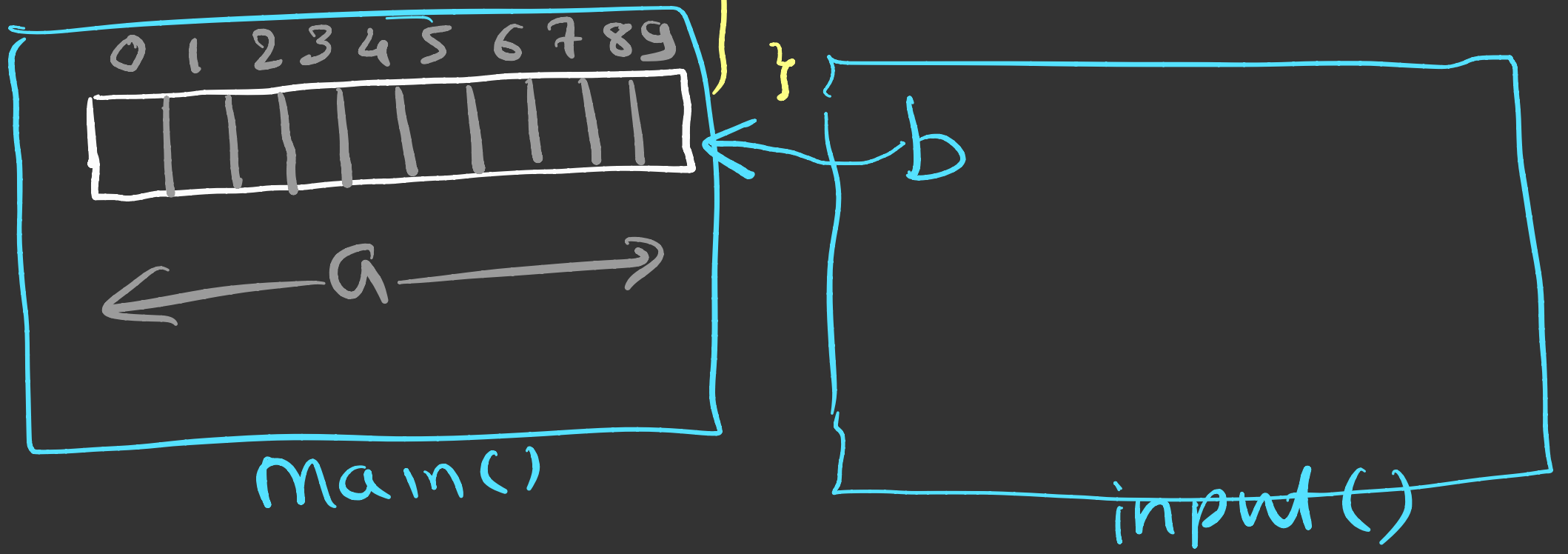
example

	0	1	2	3	4	5	6	7
given array →	20	50	90	60	70	80	30	10
Sorted array →	10	20	30	50	60	70	80	90

Function call by passing array

```
main()
{
    int a[10];
    input(a);
}
```

```
void input(int b[])
{
    int i;
    printf("Enter 10 numbers");
    for(i=0; i<=9; i++)
        scanf("%d", &b[i]);
}
```



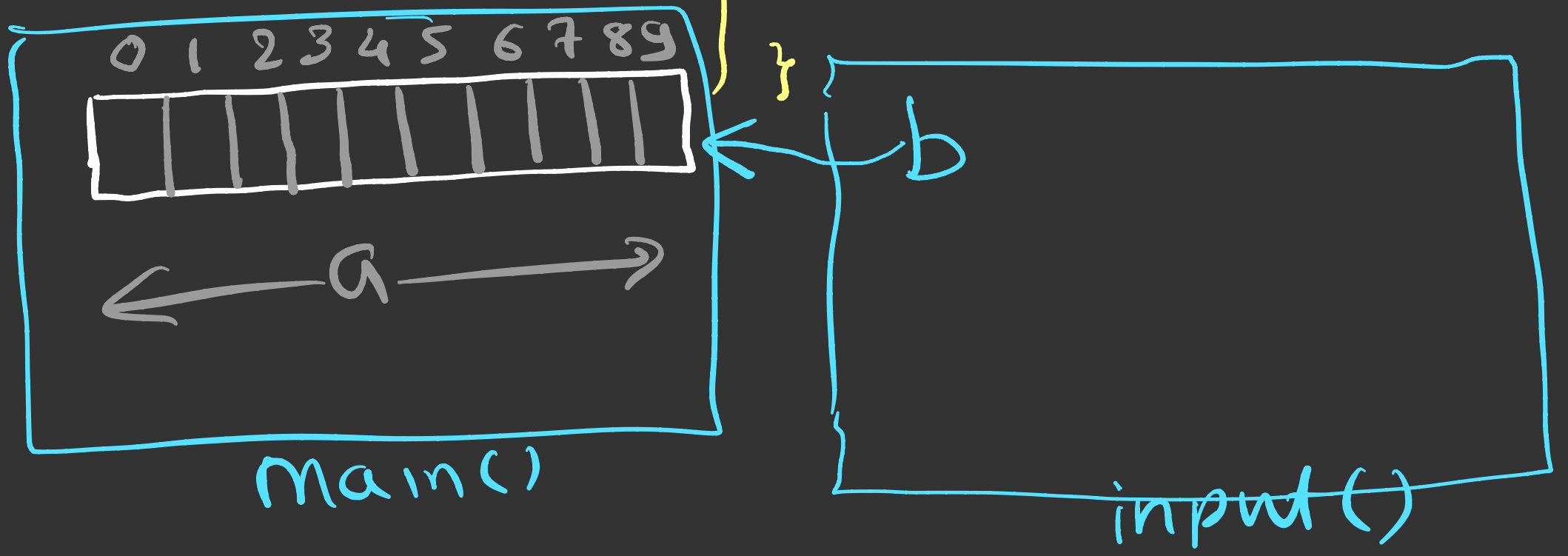
Function call by passing array

```
main()
{
    int a[5];

    input(a,5);

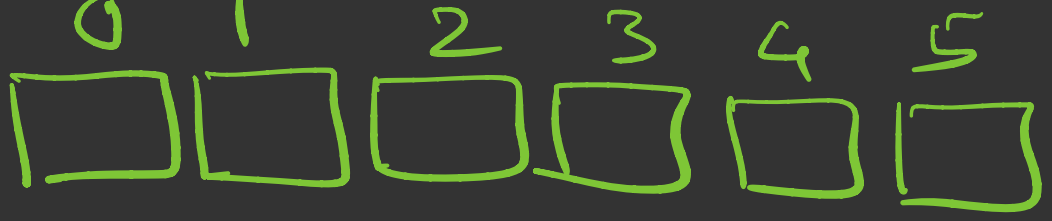
}
```

```
void input( int b[],int N)
{
    int i;
    printf("Enter %d numbers", N);
    for( i=0 ; i<N ; i++)
        scanf("%d",&b[i])
}
```



Two dimensional Arrays

```
int a[6];
```



one dimensional Array

```
int b[2][3];
```



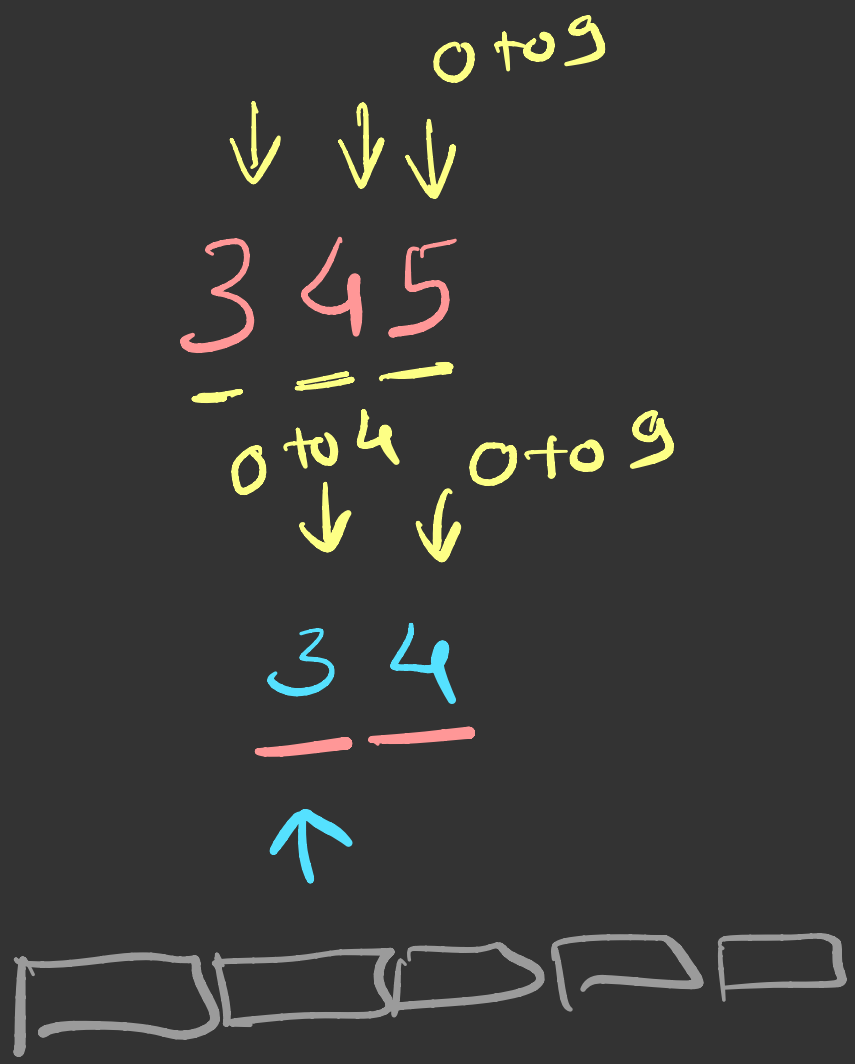
Two dimensional

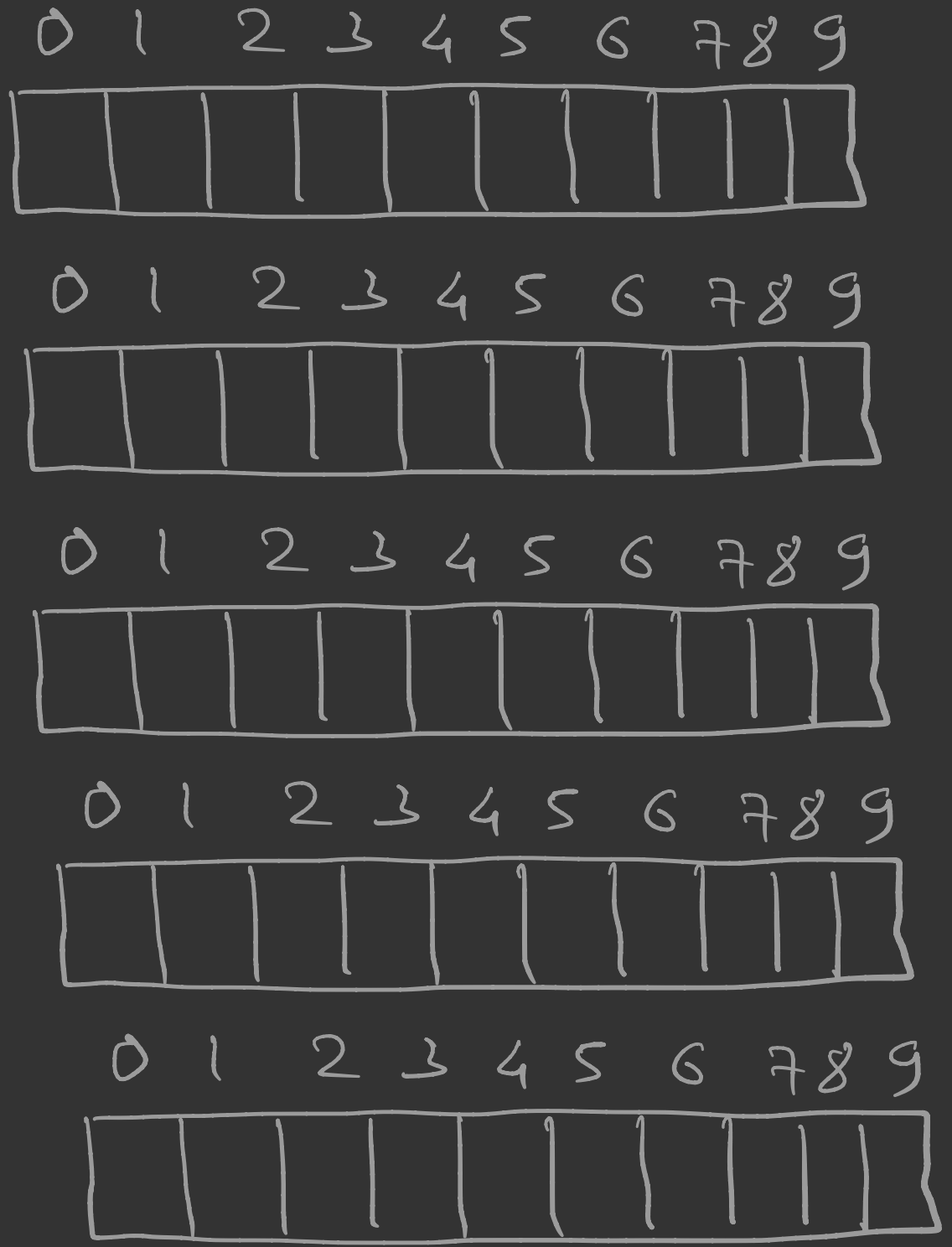
```
int c[4][5][3];
```

Three dimensional

int a[50]; ✗

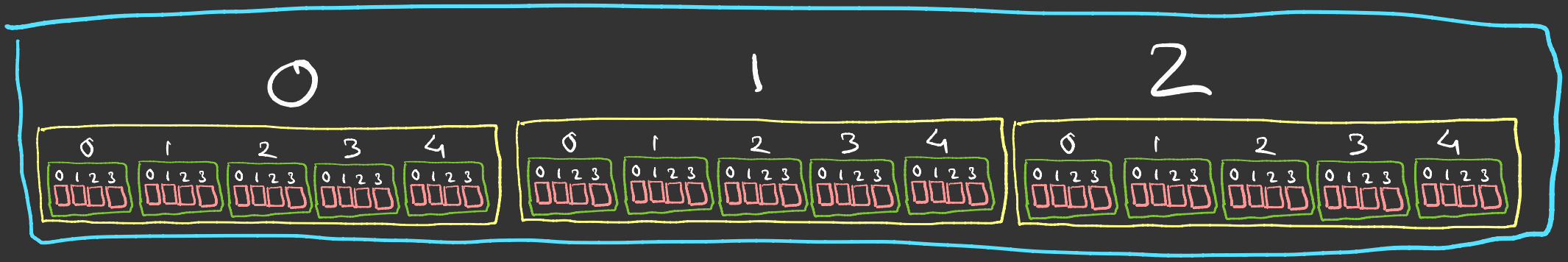
int a[5][10];





Multi-Dimensional Arrays

```
int a[3][5][4];
```




```
a[1][3][2]
```

```
int a[3][4];
int i, j;

for(i=0 ; i<=2; i++)
{
        for( j=0 ; j<=3 ; j++)
                scanf("%d", &a[i][j]);

}
```

i	j
0	0
0	1
0	2
0	3
1	0
1	1
1	2
1	3
2	0
2	1
2	2
2	3